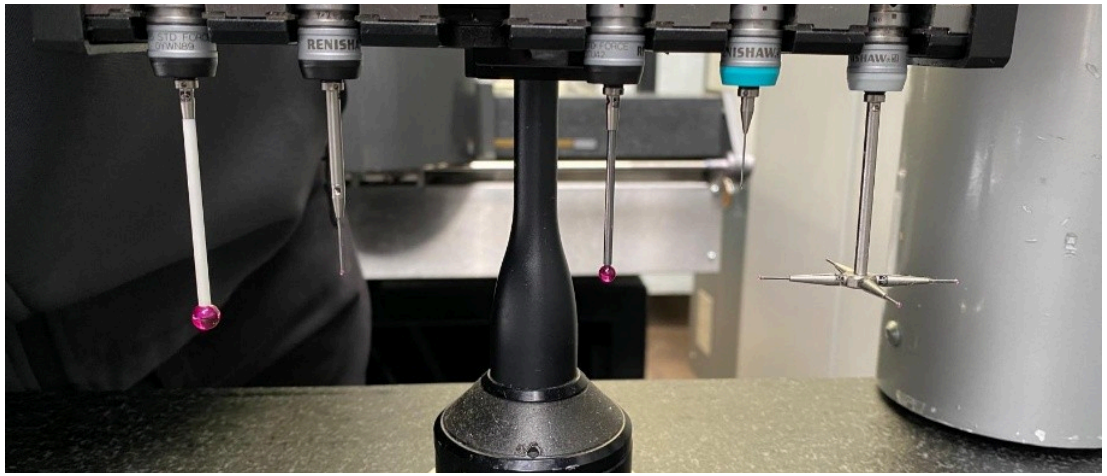
01 CMM PROGRAMMING

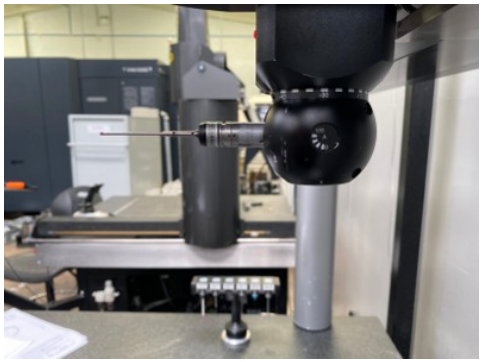
Metrology workflow from stylus selection to coordinate-system establishment

ROLE

CMM PROGRAMMER



01 STYLUS SELECTION - Ball diameter and stem length balance feature access, clearance, reach and measurement stability; where practical, the configuration is kept short and stiff.

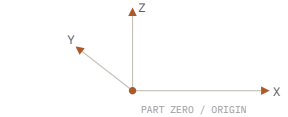


02 PH10T PROBE HEAD - Motorised indexing head for repeatable probe reorientation.

CMM COORDINATE SYSTEM

Machine coordinates define the CMM's Cartesian X, Y and Z reference frame. During programming, measured features are used to establish a **part coordinate system** aligned to the component's datum structure.

The part coordinate system then becomes the reference for subsequent feature measurement and GD&T evaluation.



CARTESIAN REFERENCE FRAME

PART ALIGNMENT & DATUM ESTABLISHMENT

The component is aligned (when applicable) to the **engineering drawing datum structure**: primary plane → secondary feature → tertiary / rotational reference. This establishes the **part coordinate system** used for subsequent feature measurement and GD&T evaluation.

PRIMARY PLANE → SECONDARY FEATURE → TERTIARY / ROTATIONAL REF. → PART COORDINATE SYSTEM

PH10T A 0°–105° · B −180° to +180° · 7.5° steps · 720 positions

GD&T REFERENCE

FORM	FORM	FORM	FORM	ORIENT.	ORIENT.	ORIENT.	LOCATION	PROFILE	PROFILE	RUNOUT	RUNOUT
Straightness	Flatness	Circularity	Cylindricity	Parallelism	Perpendicularity	Angularity	Position	Profile of a Line	Profile of a Surface	Circular Runout	Total Runout

PROCESS EXPOSURE

01 Stylus Selection → **02** Probe Qualification → **03** Part Alignment → **04** Datum Establishment → **05** Feature Measurement → **06** GD&T Evaluation

TECHNICAL EXPOSURE

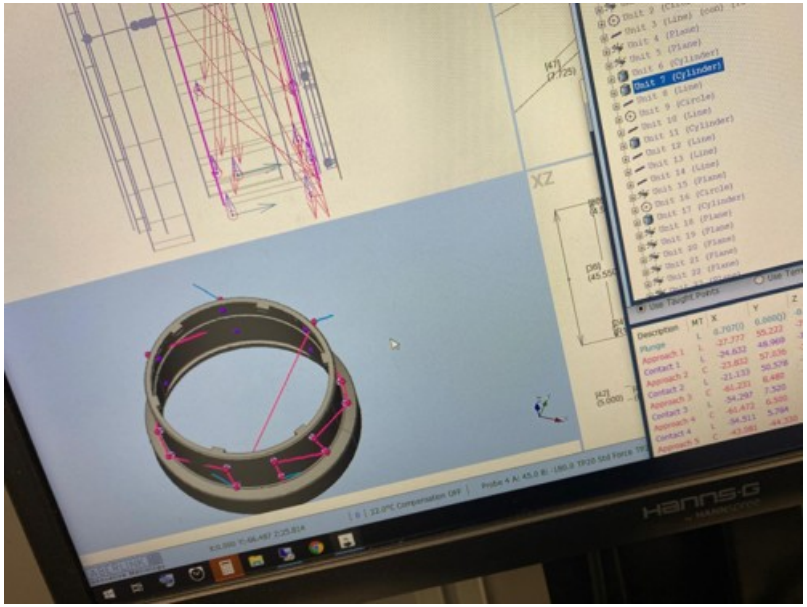
CMM Programming | Renishaw PH10T | Stylus Selection | Datum Reference Frame | Part Alignment | XYZ Coordinate System | Probe Qualification | GD&T | Dimensional Inspection

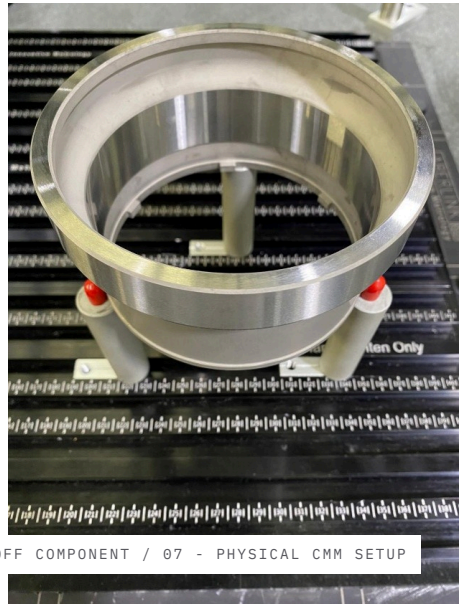
CMM PROGRAMMING

02

CAD-based inspection programming prepared before the first-off component reaches the CMM

ROLE	PROCESS	SOFTWARE	FOCUS
CMM PROGRAMMER	CMM INSPECTION	ABERLINK 3D	OFFLINE PROGRAMMING





06 - FIRST-OFF COMPONENT / 07 - PHYSICAL CMM SETUP

01 OFFLINE PROGRAMMING - CAD model, programmed features and probe-path information prepared in Aberlink 3D before physical first-off inspection

02 PHYSICAL FIRST-OFF - Component brought to the CMM for setup, online program verification and inspection

ENGINEERING CONTEXT

Offline CAD programming separates much of inspection-program development from physical CMM availability. The customer drawing establishes the inspection intent; a CAD model generated in a compatible system such as SolidWorks, AutoCAD or Siemens NX can be transferred through supported exchange formats such as STEP or IGES. In Aberlink 3D, the measurement sequence, features, probe approaches and orientations, travel and clearance strategy can then be developed before the first-off component is available.

This allows programming work to run in parallel with manufacturing, **minimising the amount of CMM programming and adjustment required after the first component arrives**. The prepared routine is then brought online and verified against the actual workholding, datum setup, probe accessibility, physical clearance, machine/probe behaviour and measurement results.

RIGHT-FIRST-TIME OBJECTIVE

Offline programming supports a right-first-time objective by developing the **CMM measurement strategy** before physical execution. Path review can reduce the risk of accessibility and movement issues, but it does not replace verification on the actual CMM.

Complex geometry can increase feature count, probe orientations, path complexity and clearance demands; however, datum structure, workholding, probe selection, accessibility and actual first-off variation also influence successful validation.

OFFLINE PROGRAM → PHYSICAL SETUP → ONLINE VERIFICATION → VALIDATED INSPECTION ROUTINE

PROCESS EXPOSURE

01 Drawing Review → 02 CAD Preparation → 03 Offline Programming → 04 Probe-Path Planning → 05 Program Verification → 06 First-Off Inspection

TECHNICAL EXPOSURE

CMM Programming | Offline Programming | Measurement Strategy | Probe-Path Planning | Probe Accessibility | Collision Clearance | First-Off Verification | CAD-to-CMM

CMM PROGRAMMING

03

Programming strategy for probe accessibility, feature construction and efficient multi-face inspection

ROLE	PROCESS	SOFTWARE	FOCUS
CMM PROGRAMMER	CMM INSPECTION	ABERLINK 3D	MOVE VIA / SAFE VOLUME



01 **MULTI-FACE COMPONENT** - Complex geometry requiring planned probe orientations, feature access and measurement sequencing.



02 **SIDE 1 / SIDE 2 ACCESS** - Alternate orientation used where internal planes and radii could not be reached from a single setup.

Programming is more than point collection. For a long inspection routine, the sequence must coordinate datum establishment, probe accessibility, stylus strategy, feature construction, collision avoidance and efficient probe travel across multiple faces.

PROGRAMMING SCOPE

Drawing package: approximately 8 pages
Inspection routine: long, multi-face feature sequence

PROGRAMMING CHALLENGES

01 / MULTI-AXIS PROBE ACCESS

Different faces and internal features required planned **probe orientations**. PH10T Probe Angles A&B indexing and stylus configuration were selected around feature accessibility, reach, clearance and measurement requirements.

02 / DATUM & FEATURE CONSTRUCTION

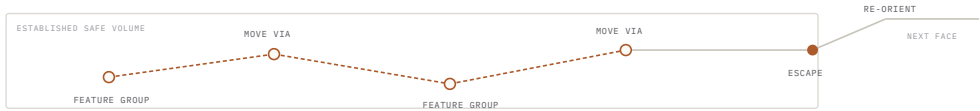
The part coordinate system was established from the required references. Additional planes were **constructed manually** so circular and arc features could be evaluated on the appropriate geometric reference.

03 / SIDE 1 ⇔ SIDE 2

Some internal planes and radii were inaccessible from one orientation, so the routine was structured into **Side 1 / Side 2** measurement sequences with controlled probe reorientation between faces.

MOVE VIA SAFE VOLUME

A deliberate path-planning layer used to reduce unnecessary rise / escape / re-entry movements while maintaining **collision avoidance**.



SAFE VOLUME CONSTRUCTION

As geometry is measured, Aberlink 3D can establish a protected **Safe Volume** around the measured component. I used strategically selected boundary / extreme features early in the routine so the relevant volume was established before detailed inspection. **Move Via** points were then introduced between measurement groups to keep travel controlled within the established volume before an intentional escape for probe reorientation or another face.

PROBE & MEASUREMENT STRATEGY

Different stylus configurations were selected according to **accessibility, reach, clearance, probe orientation and measurement performance**. Measurement-point count and distribution were treated as an engineering decision influenced by feature geometry, tolerance, form requirements and the confidence required from the inspection result.

PROCESS EXPOSURE

01 Datum Setup → 02 Probe Strategy → 03 Safe Volume → 04 Feature Planning → 05 Move Via Optimisation → 06 Multi-Face Inspection → 07 GD&T Evaluation

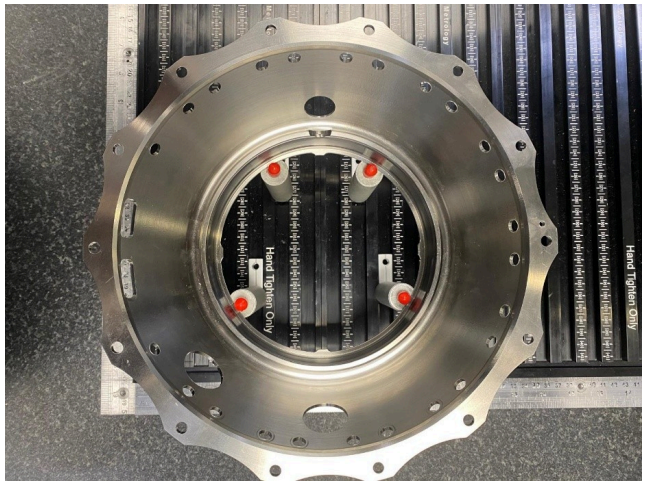
TECHNICAL EXPOSURE

CMM Programming | Aberlink 3D | PH10T | Safe Volume | Move Via | Probe Path Planning | Probe Accessibility | Stylus Selection | Datum Alignment | Feature Construction | GD&T | Inspection Optimisation

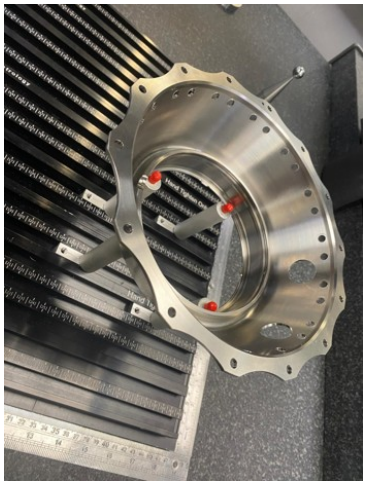
04 CMM PROGRAMMING

CAD-Based Feature Construction & Complex Geometry

ROLE	PROCESS	SOFTWARE	FOCUS
CMM PROGRAMMER	CMM INSPECTION	SOLIDWORKS + ABERLINK 3D	CAD → CMM



01 CURVED-SURFACE HOLE PATTERN - Inspection features distributed across a contoured internal surface; full drawing geometry and values intentionally omitted.



02 TWO-SIDED ACCESS - Elevated setup provided physical access to relevant underside reference features while maintaining a controlled inspection strategy.

CAD → CMM / FEATURE CONSTRUCTION

01 SOLIDWORKS MODEL → 02 DATUM / COORDINATE SYSTEM → 03 NOMINAL HOLE CENTRES · X/Y/Z → 04 ABERLINK REFERENCE POINTS → 05 CONSTRUCTED PLANES → 06 MEASURED CIRCLES → 07 PROJECT / EVALUATE

Position / profile evaluation is performed from the measured features in the established datum reference frame; CAD values remain **nominal references**, not measured results.

ENGINEERING LOGIC

The CAD model was used to derive the **nominal centre coordinates** of the hole features relative to the component datum structure. These nominal XYZ values were then used to create corresponding reference points in Aberlink 3D.

Those references supported construction of the planes needed to relate measured circles on the curved/contoured geometry to the intended inspection definition.

01 / DATUM-BASED COORDINATE SYSTEM

The inspection coordinate system was established from the drawing's **datum reference structure**. Correct alignment provides the reference frame against which positional requirements are evaluated.

02 / CONSTRUCTED REFERENCE PLANES

CAD-derived nominal points were used in Aberlink to **construct reference geometry**. This allowed measured circular features to be related to the appropriate geometric reference.

03 / HOLES ON CURVED GEOMETRY

The holes were measured as **circles** and related/projected to the constructed planes rather than treating the contoured surface as inherently planar.

04 / POSITIONAL TOLERANCE

The hole pattern included **position requirements**; the program therefore needed correct datum referencing and nominal-location comparison without exposing drawing-specific values.

PROBE ACCESS & PATH PLANNING

Probe orientation → stylus accessibility → approach direction → feature access → collision clearance → measurement sequence. The component required physical elevation on the fixture so underside datum/reference surfaces could be reached and measured rather than relying on the granite table as a substitute reference. The curved geometry also required deliberate approach and collision-avoidance planning around the internal features.

PROCESS EXPOSURE

01 CAD Review → 02 Datum Establishment → 03 Nominal Coordinate Extraction → 04 Feature Construction → 05 Probe Access Planning → 06 Hole Measurement → 07 GD&T Evaluation

TECHNICAL EXPOSURE

CMM Programming | Aberlink 3D | SOLIDWORKS | Datum Reference Frame | CAD Nominals | XYZ Coordinates | Feature Construction | Probe Accessibility | Position Tolerance | Profile Evaluation

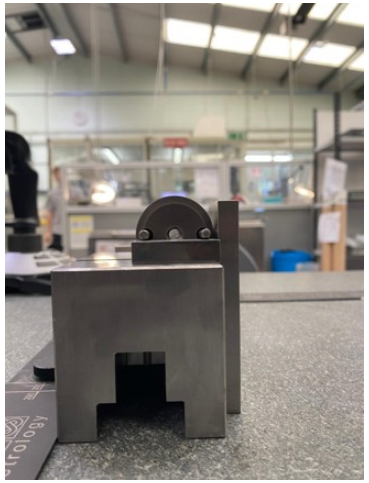
05 CMM PROGRAMMING

Repeatable Part Fixturing & Alignment

ROLE	PROCESS	SETUP	FOCUS
CMM PROGRAMMER	CMM INSPECTION	REPEATABLE FIXTURING	PART ALIGNMENT



01 **LOCATING / CLOCKING PINS** - Existing holes are used with locating elements to provide repeatable physical orientation before the CMM routine begins.



02 **REPEATABLE SETUP** - Gauge-block reference, locating pins and controlled clamping work together to reproduce the intended loading condition.

REPEATABLE SETUP / PHYSICAL LOCATION

POSITIONAL REFERENCE	→	ROTATIONAL REFERENCE	→	SECURE WORKHOLDING	→	CONTROLLED LOADING
GAUGE BLOCK		LOCATING PINS		Vice		REPEATABLE SETUP

Physical workpiece location is established by the setup arrangement; the CMM software coordinate-system alignment remains a separate inspection-programming step.

OPERATOR SETUP STANDARDISATION

The arrangement was incorporated into operator instructions so different operators could reproduce the same loading condition before running the inspection program.

The objective is **repeatability from one loading to the next**, reducing operator-dependent variation before measurement begins.

01 / REPEATABLE PART LOCATION

Existing holes are used with **locating / clocking pins** to control the component's physical orientation during loading.

02 / GAUGE-BLOCK REFERENCE

A **gauge block / slip gauge** provides a physical setup reference for reproducing the required relationship between the component feature and Vice edge.

03 / ROTATIONAL ALIGNMENT

Gauge block → positional reference; **locating pins → rotational reference**; **Vice → secure workholding**.

04 / OPERATOR STANDARDISATION

Setup instructions translate the physical arrangement into a **repeatable loading condition** before the CMM program is run.

CMM FIXTURING METROLOGY CONTEXT

Effective workholding balances **repeatability, stability, accessibility, minimal part distortion and probe clearance**. Metal locating elements can be appropriate when contact is controlled and the fit, contact surfaces and constraint strategy are suitable; clamping must avoid excessive force or unnecessary constraint that could deform the component.

REPEATABILITY | STABILITY | ACCESSIBILITY | MINIMAL DISTORTION | PROBE CLEARANCE

PROCESS EXPOSURE

01 Part Loading → **02** Gauge-Block Reference → **03** Pin Location → **04** Rotational Alignment → **05** Vice Clamping → **06** CMM Inspection

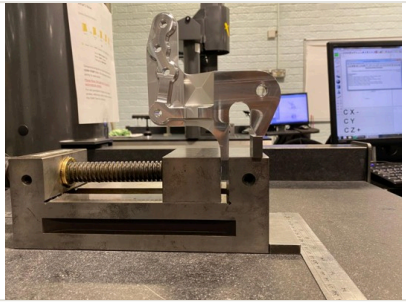
TECHNICAL EXPOSURE

CMM Fixturing | Repeatable Workholding | Locating Pins | Clocking | Gauge Blocks | Slip Gauges | Datum Location | Rotational Alignment | Fixture Repeatability | Probe Accessibility

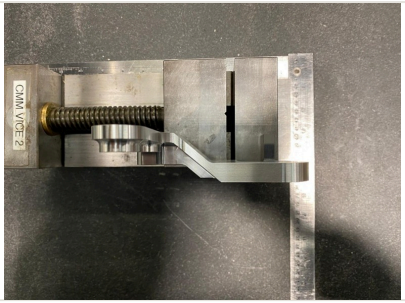
06 CMM PROGRAMMING

Minimal-Contact Fixturing & Probe Access

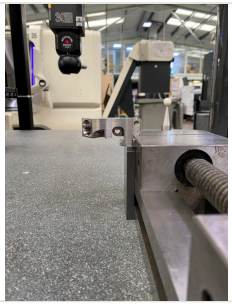
ROLE	PROCESS	SETUP	FOCUS
CMM PROGRAMMER	CMM INSPECTION	MINIMAL CONTACT	PROBE ACCESS



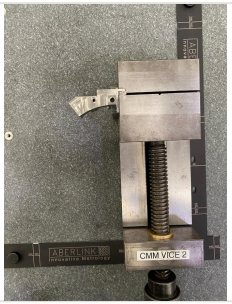
01 LARGE COMPONENT / LARGE vice - Contact is concentrated at controlled regions while the remaining geometry stays available to the probe.



02 SETUP REFERENCE - Gauge blocks / slip gauges are used as temporary physical references during setup before the inspection routine is run.



03 COMPOUND-ANGLE FEATURE - The far-side hole is accessed with a suitable probe orientation and approach direction while maintaining collision clearance.



04 PROBE ACCESS - The top view shows the component largely clear of the vice, preserving access to the required measurement surfaces and features.

FIXTURE AS PART OF THE INSPECTION STRATEGY

FEATURE REQUIREMENTS → PROBE ACCESS → FIXTURE CONTACT → CLAMPING → PROBE PATH → INSPECTION SEQUENCE

Single-setup objective: where feature accessibility permits, measure as many required features as possible without removing, reorienting or re-fixturing the component.

This can reduce operator intervention and setup time while keeping the inspection sequence continuous.

01 / MINIMAL-CONTACT FIXTURING

Workholding should provide **stability and repeatability** without unnecessarily obstructing the measurement envelope or probe path.

02 / SINGLE-SETUP INSPECTION

Programming is planned around **feature accessibility, probe orientation and measurement sequencing** where a continuous setup is practical.

03 / GAUGE BLOCKS / SLIP GAUGES

Used as temporary setup/reference aids to establish the intended position before clamping; dimensions and grades are intentionally omitted.

04 / COMPOUND-ANGLE FEATURE ACCESS

The oblique feature requires suitable **probe orientation, approach vector and collision clearance** for datum-based positional evaluation.

LARGE PART ⇔ SMALL PART SAME ACCESSIBILITY PRINCIPLE

Fixture scale changes with component size and geometry. The inspection objective does not: **secure the workpiece while maximising measurable surface access**. Physical fixturing remains distinct from the CMM software coordinate-system / datum alignment.

PROCESS EXPOSURE

01 Drawing Review → 02 Feature Accessibility → 03 Fixture Selection → 04 Minimal-Contact Setup → 05 Probe Access Planning → 06 CMM Inspection

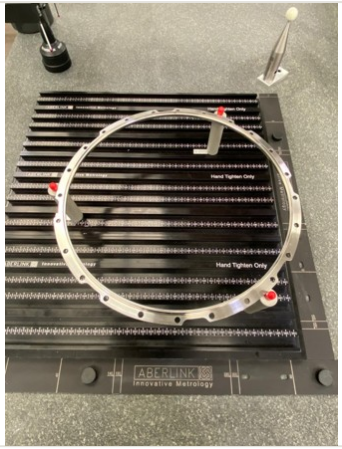
TECHNICAL EXPOSURE

CMM Programming | Minimal-Contact Fixturing | Probe Accessibility | Feature Accessibility | Fixture Repeatability | Gauge Blocks | Probe Orientation | Approach Vector | Collision Clearance | Positional Tolerance | Datum Reference Frame | Inspection Optimisation

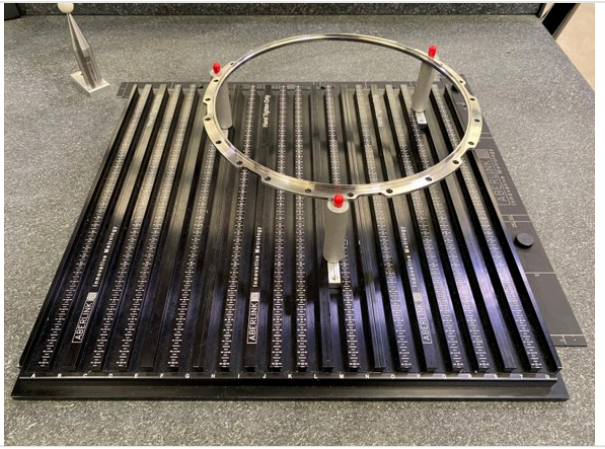
07 CMM PROGRAMMING

Fixture Validation & Measurement Repeatability

ROLE	PROCESS	SOFTWARE	FOCUS
CMM PROGRAMMER	CMM INSPECTION	ABERLINK 3D	FIXTURE VALIDATION



01 **CONTROLLED WORKHOLDING** - The ring is elevated above the fixture plate so the underside remains accessible while locating points provide controlled restraint.



02 **RESTRAINT VS ACCESS** - Locator placement must stabilise the workpiece without unnecessarily constraining the circular geometry or obstructing probe access.

FIXTURE VALIDATION & MEASUREMENT REPEATABILITY

The objective was not maximum clamping force. The workholding condition had to prevent movement under probing while avoiding excessive constraint that could alter measured geometry. Repeated loading and measurement trials were used to assess setup behaviour and refine locator positions/restraint before documenting the loading condition for operators.

EXCESSIVE CONSTRAINT
Clamping-induced distortion → altered workpiece geometry.

→

OPTIMISED CONSTRAINT
Stable workpiece → preserved geometry → repeatable measurement.

→

INSUFFICIENT CONSTRAINT
Workpiece slippage under probing force → point-to-point displacement → reduced repeatability.

01 / PROBE-INDUCED DISPLACEMENT

When a circular feature is acquired from multiple touch points, the CMM calculates the feature from those coordinates. If the workpiece moves between contacts, the points may no longer represent one stationary physical position.

02 / POSITION & CIRCULARITY

For holes with **position tolerances**, movement can affect the reported feature location relative to the established datum reference frame. **Circularity** remains a separate form control.

03 / CONTROLLED CONTACT

Locator position, contact condition and workpiece elevation were treated as part of the **measurement strategy**, balancing stability, accessibility and the risk of clamping-induced distortion.

04 / SETUP REPRODUCIBILITY

The selected arrangement was documented through **photographs and operator instructions** so the component could be loaded consistently without claiming perfect repeatability or eliminating all variation.

PROCESS EXPOSURE

01 Fixture Review → 02 Locator Positioning → 03 Controlled Loading → 04 Repeatability Trial → 05 CMM Measurement → 06 Fixture Optimisation → 07 Operator Work Instruction

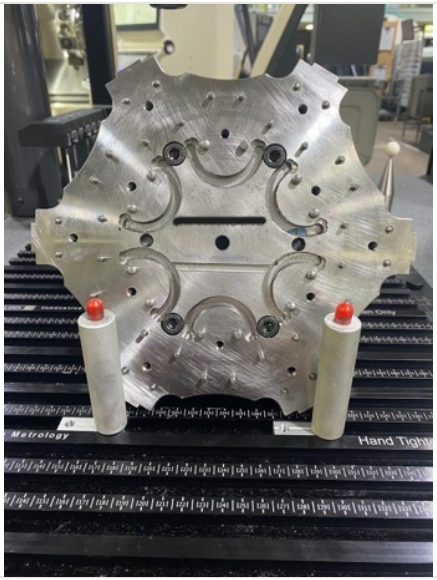
TECHNICAL EXPOSURE

CMM Programming | Fixture Validation | Workpiece Constraint | Loading Repeatability | Probe-Induced Displacement | Clamping-Induced Distortion | Datum Stability | Position Tolerance | Circularity | GD&T Verification

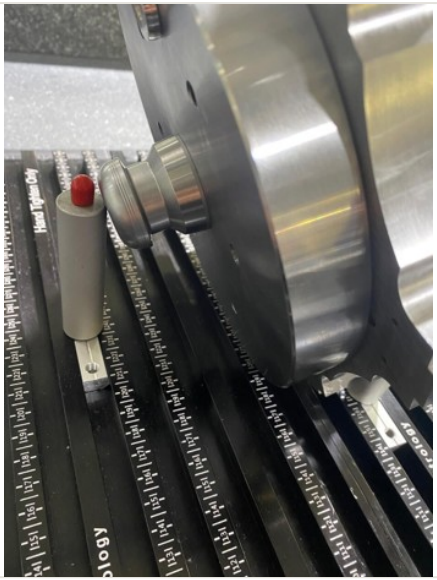
08 CMM PROGRAMMING

Fixture Metrology & CAD/CAM Process Feedback

ROLE	PROCESS	OUTPUT	FEEDBACK
CMM PROGRAMMER	FIXTURE METROLOGY	ABERLINK DATA	CAD / CAM



01 **MULTI-PART FIXTURE** - Locating features and fasteners position multiple component locations for a machining cycle. The fixture is treated here as a **measurement object**.



02 **FIXTURE GEOMETRY** - The CMM captures measured relationships between locator positions, hole locations, diameters, orientations and other relevant reference features.

CMM → CAD / CAM / MANUFACTURING FEEDBACK

PHYSICAL FIXTURE → CMM MEASUREMENT → ACTUAL FEATURE LOCATIONS → ABERLINK DRAWING / MEASUREMENT DATA → CAD / CAM DEVELOPMENT → CNC MACHINING

Measured fixture geometry was supplied as engineering information to support the machining setup and CAD/CAM program development. CMM measurement did not directly control the CNC machine.

SIX-PART FIXTURE / BATCH MANUFACTURING

One fixture → six component locations → one CNC cycle.

The inspection strategy captured the relevant fixture geometry across the component locations so the machining team could work from measured positional relationships.

01 / FIXTURE METROLOGY

Measure **locator position, hole locations, feature diameters, angular relationships, orientations and feature-to-feature distances** without inventing numerical values.

02 / COORDINATE FRAME

Establish the measured spatial relationship between the fixture's reference geometry and locating features; keep **measured geometry** distinct from nominal CAD.

03 / ABERLINK OUTPUT

Detailed **Aberlink drawings / measurement outputs** documented relevant distances, centres, diameters, angles, orientations and feature relationships for the machining team.

04 / MANUFACTURING FEEDBACK

The measured information supported **CAM setup and feature-location decisions**, reducing uncertainty when transferring the physical fixture into the programming environment.

MEASUREMENT DATA → MANUFACTURING CONFIDENCE

Accurate fixture measurement supported the six-axis milling team leader in developing the CAD/CAM machining setup. The objective was to improve the likelihood of **right-first-time machining** by reducing uncertainty in the measured fixture geometry - not by claiming that CMM measurement alone guaranteed first-pass acceptance.

PROCESS EXPOSURE

01 Fixture Review → 02 Datum Establishment → 03 Locator Measurement → 04 Feature Verification → 05 Aberlink Documentation → 06 CAD/CAM Feedback

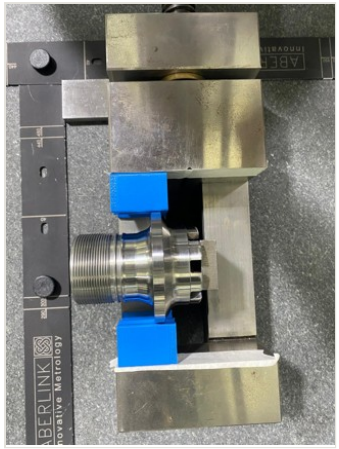
TECHNICAL EXPOSURE

Fixture Metrology | CMM Programming | Aberlink 3D | Datum Reference | Coordinate Frames | Locator Position | Feature Location | Dimensional Verification | CAD/CAM Integration | Manufacturing Feedback

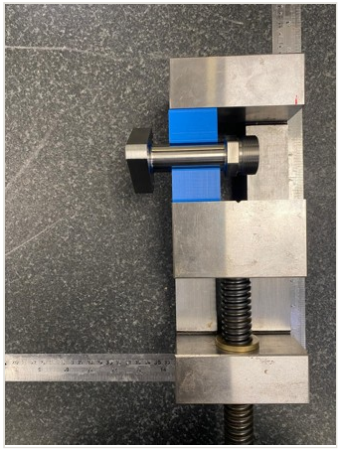
09 CMM PROGRAMMING

Custom 3D-printed PLA workholding for repeatable setup, probe access and surface protection

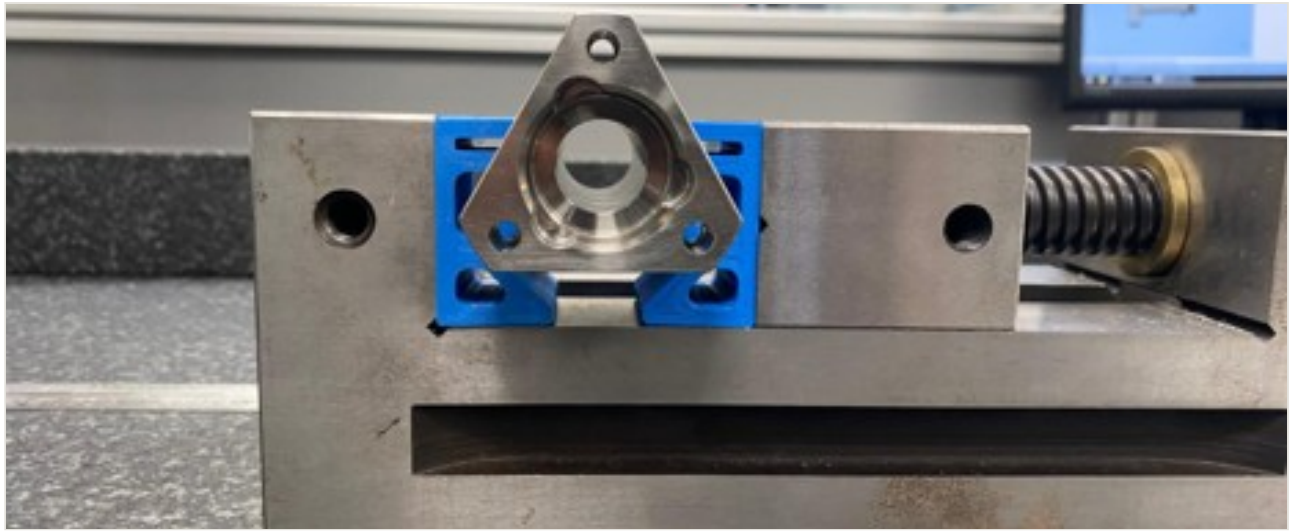
ROLE	PROCESS	SOFTWARE	FOCUS
CMM PROGRAMMER	CMM INSPECTION	ABERLINK 3D	ADDITIVE MANUFACTURING



01 **Component A - Top View**



02 **Component B - Top View**



02 **Component B - Front View** Front view of the same Setup, showing 3D-printed brackets used to clamp the component's shaft into position.

ENGINEERING CONTEXT

Custom FDM 3D-printed PLA fixtures were used as practical metrology tooling to position components consistently during CMM inspection. The supports were shaped around the component geometry so that the part could be loaded repeatably without unnecessarily covering inspection surfaces.

For CMM work, the fixture is part of the measurement setup: a stable, repeatable location reduces setup variation, while open support geometry preserves probe access. The objective is controlled restraint rather than simply maximising clamping force.

METROLOGY DESIGN CONSIDERATIONS

01 **Repeatable location** - define a consistent loading position so the inspection routine starts from the same physical condition.

02 **Constraint control** - locate securely without over-constraining a part that could deflect or distort under restraint.

03 **Probe access** - keep contact points away from measured surfaces and preserve practical probe approach paths.

04 **Fixture validation** - confirm fit, seating and repeatable loading before relying on the fixture for inspection.

FIXTURE LOGIC

PART GEOMETRY

↓

FIXTURE GEOMETRY

↓

CONTROLLED WORKPIECE LOCATION

↓

PROBE ACCESSIBILITY

↓

REPEATABLE CMM MEASUREMENT

WHY FDM PLA IS USEFUL HERE

01 **Geometry freedom** - FDM can produce customised support shapes without conventional machining of complex contours.

02 **Rapid iteration** - fixture geometry can be modified and reprinted as the component or inspection requirement changes.

03 **Low-mass tooling** - polymer supports are practical for lightweight inspection workholding and quick handling.

04 **Process sensitivity** - PLA print accuracy depends on parameters such as layer height, orientation and thermal conditions, so the fixture should be checked after manufacture.

05 **Inspection-focused design** - the value comes from repeatable positioning and access to the measurement surfaces, not simply from holding the part tightly.

PROCESS EXPOSURE - FROM FIXTURE DESIGN TO INSPECTION

01 Component Review → 02 Fixture Design → 03 3D Print (PLA) → 04 Setup & Validate → 05 CMM Measurement → 06 Operator Instruction

TECHNICAL EXPOSURE

CMM Programming | Additive Manufacturing | FDM / PLA | Custom Fixturing | Probe Accessibility | Loading Repeatability | Surface Protection | Fixture Validation | Metrology Workholding

TECHNICAL BASIS: Renishaw metrology fixturing guidance · ZEISS workpiece-fixturing guidance · NIST additive-manufacturing guidance · peer-reviewed FDM/PLA dimensional-accuracy studies

10

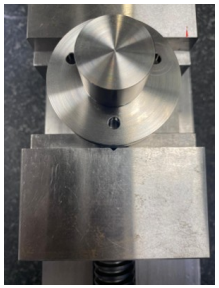
CMM PROGRAMMING

Three Classes of Aerospace Grade CMM Programmable Parts

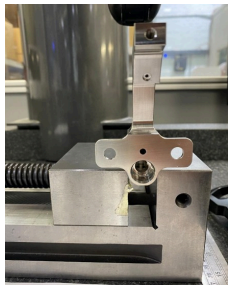
ROLE	PROCESS	SOFTWARE	FOCUS
CMM PROGRAMMER	CMM INSPECTION	ABERLINK 3D	COMPONENT CATEGORY



01 LARGE / COMPLEX - ANNULAR COMPONENT - Cylindrical, planar and hole-pattern features requiring a geometry-driven measurement strategy



02 MEDIUM - ROTATIONAL COMPONENT - Circular, cylindrical and hole-pattern features for feature-based inspection



03 SMALL / FEATURE-DENSE - COMPACT COMPONENT - Multiple orientations and restricted access can drive programming complexity

01 LARGE / COMPLEX → **02** MEDIUM / ROTATIONAL → **03** SMALL / FEATURE-DENSE

RANGE SIZE ≠ PROGRAMMING COMPLEXITY

RANGE OF INSPECTION GEOMETRY

CMM programming required the inspection strategy to be adapted to the component's geometry, feature density and accessibility. Across large annular components, rotational parts and compact feature-dense components, programming involved establishing appropriate datum references, selecting measurable features, planning probe approaches and sequencing the inspection to obtain reliable dimensional and geometric results.

FEATURE-BASED INSPECTION

The program was structured around measurable geometric features rather than simply probe movements: **planes, cylinders, circles, holes, bores, diameters, annular surfaces, feature centres and feature-to-feature relationships.** Their evaluation was driven by the drawing and applicable GD&T requirements; the photographs show representative component geometries, not a claim that every listed feature was measured on every part.

GEOMETRY + DATUM STRUCTURE + FEATURE DENSITY + ACCESSIBILITY + PROBE STRATEGY

PROCESS EXPOSURE

01 Drawing Interpretation → **02** Datum Strategy → **03** Feature Selection → **04** Probe Planning → **05** CMM Programming → **06** Feature Evaluation → **07** GD&T Verification

TECHNICAL EXPOSURE

CMM Programming | Feature-Based Inspection | Datum Reference Frames | Cylindrical Features | Hole Patterns | Probe Accessibility | Measurement Strategy | GD&T Verification | Feature Evaluation